Modeling Student Interactions in Online Statistics Courses

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Research Goals

- Analyze previously coded Collaborative Keys from online STAT 217 Courses from Winter and Fall of 2023 at Cal Poly.
- Adapt cognitive presence coding schemes to align with the Collaborative Key structure.

Collaborative Keys (CKs)

7. Do you think asking each of you to take 20 words instead of 10 words would have helped with this issue? Explain.

INITIAL ANSWERS

Student 1: I do not think that taking more words would affect the issue of bias since the means would simply show the bias more.

Student 2: I think asking to choose 20 words instead of 10 will help with the long run proportion of the words chosen by the students but it wouldn't create a bias.

DISCUSSION

Student 1: It seems like we all kind of got the same answer, but I don't know which is the correct one. What do u guys think?

Student 2: After discussing, I do think that using a higher number of words would lead to a change that would be closer to the overall mean. I think I originally did not see how an increase in statistics would change too much due to random chance.

FINAL GROUP ANSWER

Yes, with more words the student means would average out closer to the population mean of 4.2

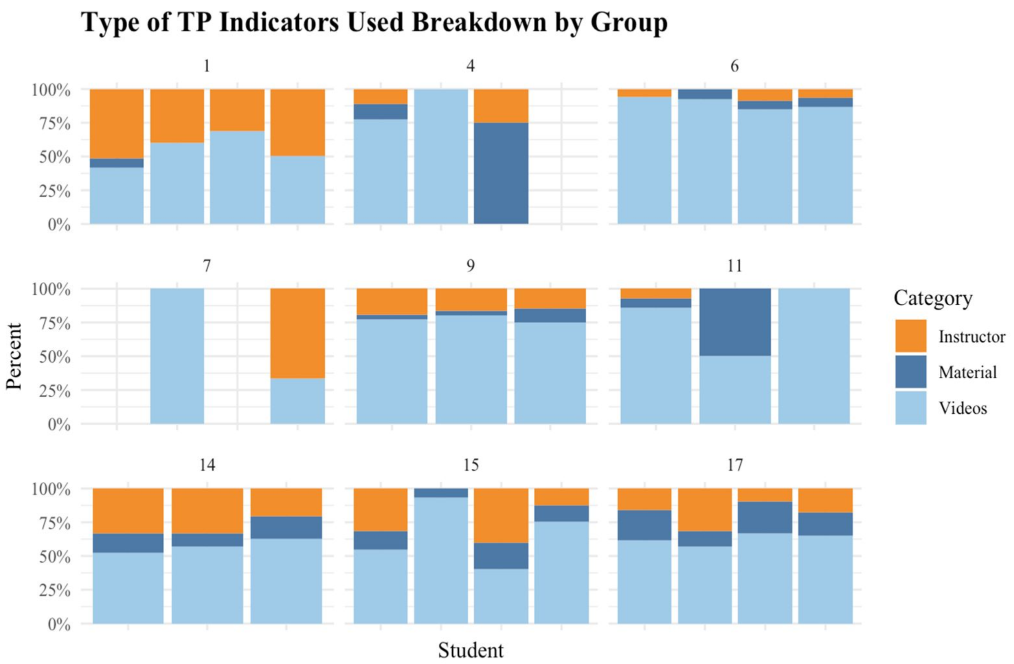
Community of Inquiry Framework

Cognitive Presence: extent of students’ learning and application of critical thinking.

Social Presence: students’ ability to present themselves as real people.

Teacher Presence: presentation of course material, and the facilitation of learning.

Teaching Presence Indicators



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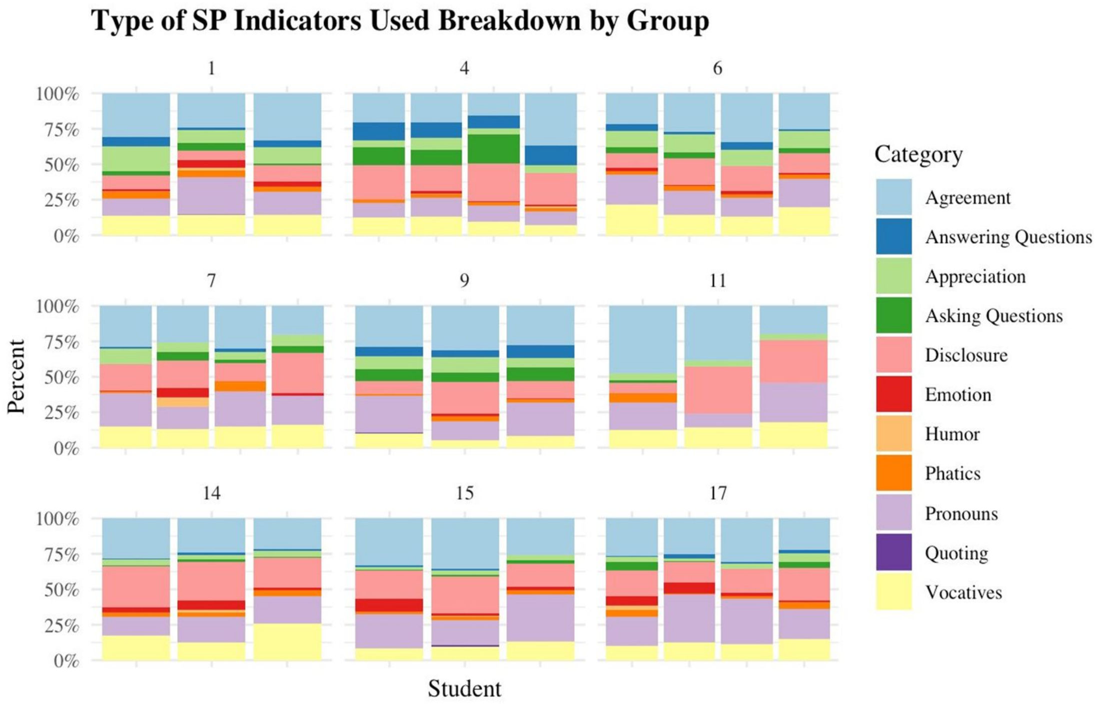
Social Presence Indicators



Lily Cook. (2025). Investigating Social Presence in Collaborative Keys for Asynchronous Courses. California Polytechnic State University.

Final Score ~ Pronouns + Asking Questions + Answering Questions

- $R^2 = 0.56$
- Positive association with final course scores: *use of inclusive pronouns* and *answering questions*
- Negative association with final course scores: *Asking questions*

Predictor	Estimate	Standard Error	T	p-value
(Intercept)	73.525	1.738	42.304	0.000
Pronouns	0.258	0.046	5.582	0.000
Asking Questions	-0.399	0.141	-2.832	0.008
Answering Questions	0.451	0.175	2.578	0.016

Coding Scheme Adaptations: Cognitive Presence

Learning Phase	Description	Adaption
Triggering Event	dissonance and unease	no adaptations

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		self-evaluation + stats

Coding Scheme Adaptations: Cognitive Presence

Learning Phase	Description	Adaption
Triggering Event	dissonance and unease	no adaptations
Exploration	searching for information and making sense of the problem	self-evaluation
		self-evaluation + stats
Integration	turning knowledge into coherent idea	peer evaluations/correction
Resolution	application of idea or hypotheses	defending solutions with own words
		defending solutions with reference to videos and slides
		selecting from initial answers

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CK Examples

- Peer Correction, Defending Solutions With Own

"Student 1: Hi Student 2, I believe the answer is D. If you look at the p-value from the graph see it is 0.000 because our statistic was 0.334 which is greater than 0.25. With that being said I think we have convincing evidence that the probability that the receiver will pick the correct target is greater than 0.25, but don't have convincing evidence that it is equal to 0.25 given that our p-value was 0.000."

Name	Peer Correction	Defending Solutions w/ Video	Defending Solutions w/ Own Words
Student 1	1	0	1

CK Examples

- Self -Evaluation + Stats, Defending Solutions With Video Reference

"Student 2: I agree^ I do think my answer had a couple of variables being mixed up in the formula, so it resulted in the incorrect answer. After watching the video wrap-up, I can confirm that it will be good for our final answer."

Name	Self Eval	Defending Sol'ns w/ Video	Defending Sol'ns w/ Own Words
Student 1	1	1	0

Acknowledgements

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